

Network Dynamics and Synaptic Plasticity in Spiking Neural Networks

Brain Modelling Course – Coding Project Report

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February 2026

1 Starting Point and Motivation

This project extends the basic Leaky Integrate-and-Fire (LIF) neuron model introduced in the course handson (`LIF_intro.ipynb`) to explore network-level phenomena in spiking neural networks using the NEST simulator. The handson provided a foundation with a single LIF neuron governed by the differential equation:

$$\tau_m \frac{dV}{dt} = -(V - E_L) + \frac{I}{g_L} \quad (1)$$

where τ_m is the membrane time constant (10 ms), V is membrane potential, E_L is resting potential (−75 mV), and I is input current. While this captures fundamental spiking dynamics with threshold-based firing ($V_{th} = -55$ mV) and reset mechanisms ($V_{reset} = -75$ mV), biological neural systems exhibit emergent properties through network interactions and synaptic plasticity that cannot be understood from single neurons alone.

2 Goals and Methodology

The project investigates three key aspects of spiking neural networks: (1) network construction with varying connectivity patterns, (2) synaptic plasticity via STDP, and (3) emergent network dynamics. We implemented a recurrent network of 100 excitatory LIF neurons using NEST simulator version 3.9 with sparse random connectivity (probability $p = 0.1$, yielding approximately 986 connections).

To achieve physiologically realistic firing rates (target: 50–100 Hz), network parameters were tuned through systematic exploration. Initial simulations with standard parameters (800 Hz Poisson input, 50 pA synaptic weights) produced silent networks due to insufficient drive relative to the sparse connectivity ($p = 0.1$ yields only ~ 10 inputs per neuron). The external drive was increased to 1500 Hz with 80 pA weights to compensate for weak recurrent coupling in the sparse regime, ensuring all neurons reach spiking threshold. This configuration mimics cortical networks where sparse local connectivity is balanced by strong external input from thalamus and other cortical areas. Recurrent synaptic weights were maintained at 20 pA to preserve sparse coupling dynamics while allowing STDP-driven weight evolution.

STDP learning modifies synaptic weights based on spike timing according to:

$$\Delta w(\Delta t) = \begin{cases} A_+ \cdot e^{-\Delta t/\tau_+} & \text{if } \Delta t > 0 \text{ (pre before post)} \\ A_- \cdot e^{\Delta t/\tau_-} & \text{if } \Delta t < 0 \text{ (post before pre)} \end{cases} \quad (2)$$

where $\Delta t = t_{post} - t_{pre}$. Implementation used NEST’s built-in `stdp_synapse` with $\tau_+ = 20$ ms, initial weights of 20 pA, and maximum weights capped at 100 pA to prevent runaway potentiation.

3 Results and Outcomes

3.1 Network Activity Patterns

The sparse recurrent network exhibited asynchronous irregular activity characteristic of cortical networks. With 100 neurons over 1000 ms, the network produced 6830 total spikes, yielding a mean population firing rate of 68.30 Hz with all 100 neurons active. The raster plot (Figure 1 in notebook) demonstrates spatially and temporally distributed spiking without strong synchronization patterns, indicating a balanced asynchronous regime. Membrane potential traces from sample neurons show irregular sub-threshold fluctuations driven by sparse synaptic input, with spikes occurring when integrated input crosses threshold.

3.2 STDP-Driven Weight Evolution

Over 1000 ms of simulation with STDP enabled, synaptic weights evolved dramatically from their initial uniform value of 20.0 pA to a final mean of 49.41 pA, representing a 147% increase. The distribution of weight changes shows a narrow spread (standard deviation 1.31 pA), centered at +29.41 pA, indicating systematic potentiation rather than bidirectional plasticity. This asymmetric weight evolution suggests that the network activity pattern—with mean firing rate 89.73 Hz during STDP simulation—favors causal spike timing relationships (pre-before-post) that drive long-term potentiation. The final weight distribution (Figure 2) peaks around 49 pA with minimal variance, demonstrating network-wide strengthening of recurrent connections.

3.3 Connectivity Effects

Comparing sparse ($p = 0.1$) and dense ($p = 0.3$) connectivity revealed significant dynamical differences. Dense networks showed 58.7% higher population firing rates (108.37 Hz vs 68.30 Hz), consistent with increased recurrent excitation. Visual inspection of raster plots shows denser networks maintain higher baseline activity with more uniform firing across neurons. Population activity fluctuations (Figure 3) demonstrate that dense networks sustain consistently elevated rates around 108 Hz, while sparse networks exhibit more variability around their 68 Hz baseline, reflecting weaker recurrent coupling and stronger influence of external input fluctuations.

3.4 Activity Pattern Analysis

Network-level dynamics were characterized by extending the coefficient of variation (CV) analysis from the handson to the network scale. Results show that CV of inter-spike intervals decreased substantially from 0.332 (sparse) to 0.182 (dense), indicating more regular individual neuron firing in densely connected networks. This suggests dense connectivity regularizes firing through stronger mean-field drive while maintaining asynchronous population activity. Analysis of population spike count distributions (Figure 4) shows dense networks have narrower, more concentrated distributions, confirming reduced temporal variability. The variance-to-mean ratio of binned spike counts was approximately 1.0 for both network types, indicating Poisson-like population statistics despite differences in individual neuron regularity.

4 Mathematical Frameworks Applied

The project employed four complementary frameworks to analyze network dynamics:

(1) LIF Neuron Model: The Leaky Integrate-and-Fire model provides a simplified but biologically motivated description of neuronal spiking. Neurons integrate synaptic input until reaching threshold ($V_{th} = -55$ mV), then fire and reset ($V_{reset} = -75$ mV) with a refractory period (2 ms). Key parameters include membrane time constant ($\tau_m = 10$ ms) and capacitance ($C_m = 250$ pF), which determine integration timescales.

(2) STDP Learning Rule: Spike-Timing-Dependent Plasticity implements Hebbian learning with temporal precision. Synapses strengthen when pre-synaptic spikes consistently precede post-synaptic spikes (causal relationship), and weaken for anti-causal timing. The exponential time windows ($\tau_+ = 20$ ms) determine the temporal sensitivity of learning, allowing the network to capture spike-timing correlations on the timescale of tens of milliseconds.

(3) Network Connectivity: Random sparse connections were created by connecting each neuron pair with independent probability p . For $p = 0.1$ with 100 neurons, this results in approximately 1000 expected connections (out of 10,000 possible pairs), with actual implementations varying slightly due to stochastic sampling (986 connections observed in our simulation). This sparse connectivity mimics biological cortical networks where neurons connect to only a small fraction of nearby cells.

(4) Statistical Measures: Building on the coefficient of variation (CV) introduced in the handson for single-neuron regularity analysis, we extended this metric to characterize network-level dynamics. Network activity was quantified using: (i) population firing rate (total spikes per neuron per second) measuring average network activity, (ii) CV of inter-spike intervals computed across all neurons to measure firing regularity, and (iii) variance-to-mean ratio of binned population spike counts to characterize temporal fluctuations in network-wide activity.

5 Discussion and Conclusions

This project successfully demonstrated that network-level dynamics emerge from the interplay of neuron properties, synaptic weights, and topology. Key findings include: (1) sparse recurrent connectivity produces asynchronous irregular states resembling *in vivo* cortical recordings, (2) STDP drives systematic weight strengthening when network activity favors causal spike relationships, leading to 147% weight increases over 1 second, (3) connectivity density strongly modulates firing rates (59% increase) and regularity (CV reduction by 45%) while maintaining asynchronous population activity, and (4) extending the CV analysis from single neurons (as in the handson) to networks reveals how connectivity shapes firing pattern regularity.

An important methodological note: initial simulations with standard parameters (800 Hz Poisson, 50 pA weights) yielded silent networks, requiring parameter adjustment to achieve physiological firing rates. This illustrates a fundamental principle in spiking network modeling: the activation regime depends on the ratio of external drive to recurrent coupling strength. In sparse networks ($p = 0.1$), each neuron receives only ~ 10 recurrent inputs, making external drive the dominant source of depolarization. The parameter adjustment (1500 Hz, 80 pA) compensates for this sparse connectivity by providing sufficient input current ($I_{ext} \approx 1500 \times 80 / 1000 = 120$ pA per neuron on average) to overcome leak conductance and reach threshold. This demonstrates that network activation regimes depend critically on the balance between external drive, recurrent coupling strength, and connectivity sparsity, a relationship absent in single-neuron models.